

CHIFENG, China

WMO: 542180

Lat: 42.3080N

Lon: 118.8325E

Elev: 656

StdP: 93.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.5	-18.7	-30.8	0.2	-15.5	-28.8	0.3	-14.6	9.1	-7.1	8.1	-6.8	1.3	220	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	33.1	19.5	31.5	19.2	30.0	18.7	22.9	29.1	22.0	27.6	21.1	26.5	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	17.0	26.1	20.2	16.1	25.2	19.2	15.2	24.4	71.7	29.0	67.9	27.5	64.6	26.6	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.2	6.2	DB	-22.8	36.0	2.3	1.8	-24.4	37.3	-25.7	38.4	-27.0	39.4	-28.7	40.7
			WB	-23.3	24.4	2.1	1.0	-24.9	25.1	-26.1	25.7	-27.3	26.3	-28.9	27.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-9.9	-6.1	0.9	9.8	16.8	21.4	23.9	22.1	16.7	8.9	-0.9	-7.8	(6)
(7)		DBStd	12.64	4.27	4.96	5.42	5.15	4.26	3.40	2.66	2.81	3.75	4.67	5.49	4.37	(7)
(8)		HDD10.0	2383	618	452	284	66	3	0	0	3	77	329	551		(8)
(9)		HDD18.3	4208	876	685	539	259	81	11	1	3	72	294	578	809	(9)
(10)		CDD10.0	1671	0	0	4	59	213	341	432	375	205	42	1	0	(10)
(11)		CDD18.3	456	0	0	0	3	32	102	174	121	24	1	0	0	(11)
(12)		CDH23.3	4298	0	0	1	53	412	971	1562	1021	264	14	0	0	(12)
(13)		CDH26.7	1481	0	0	0	11	129	352	603	329	56	1	0	0	(13)
(14)	Wind	WSAvg	2.4	2.2	2.4	3.0	3.2	3.0	2.3	2.0	1.8	2.0	2.2	2.2	2.3	(14)
(15)	Precipitation	PrecAvg	378	1	2	7	19	37	72	119	61	32	20	7	1	(15)
(16)		PrecMax	524	7	10	25	56	88	136	274	145	88	67	44	5	(16)
(17)		PrecMin	239	0	0	0	0	2	13	21	12	6	0	0	0	(17)
(18)		PrecStd	80	2	3	8	16	22	35	54	31	21	16	9	2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	10.6	20.2	28.2	33.0	34.8	35.6	33.6	30.6	25.2	16.6	7.0	(19)
(20)		MCWB	-0.9	2.5	7.9	12.5	15.7	17.3	21.5	21.0	17.8	13.3	6.7	1.2		(20)
(21)		2%	DB	1.6	7.7	16.2	24.4	30.0	32.4	33.4	31.5	28.2	22.0	12.9	3.6	(21)
(22)		MCWB	-3.2	0.7	5.9	11.1	14.6	17.8	21.0	20.4	16.2	11.3	5.3	-1.5		(22)
(23)		5%	DB	-0.4	5.2	13.1	22.0	27.6	30.5	31.7	30.0	26.4	19.5	10.2	1.5	(23)
(24)		MCWB	-4.6	-0.7	4.1	9.8	13.4	17.1	20.7	19.8	15.3	10.2	3.8	-3.0		(24)
(25)		10%	DB	-2.4	2.7	10.3	19.4	25.4	28.6	30.0	28.5	24.5	17.0	7.6	-0.5	(25)
(26)		MCWB	-6.1	-2.2	2.6	8.7	12.7	16.8	20.2	19.2	14.7	8.9	2.0	-4.4		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	3.3	8.4	14.4	17.5	21.0	24.4	24.0	20.0	14.7	7.7	1.6	(27)
(28)		MCDWB	4.3	9.1	18.5	24.2	27.9	28.5	31.5	30.7	27.7	22.0	13.5	7.3		(28)
(29)		2%	WB	-3.0	1.0	6.4	12.2	16.1	20.0	23.2	22.7	18.2	12.8	5.7	-1.4	(29)
(30)		MCDWB	1.4	7.2	15.3	22.0	26.3	27.2	29.8	28.4	24.6	19.7	12.2	3.4		(30)
(31)		5%	WB	-4.6	-0.6	4.6	10.7	15.0	19.2	22.4	21.7	17.0	11.3	4.0	-2.9	(31)
(32)		MCDWB	-0.5	5.1	12.3	20.0	24.1	26.0	28.5	27.0	23.8	17.9	9.8	1.3		(32)
(33)		10%	WB	-6.0	-2.2	2.8	9.3	13.8	18.4	21.6	20.8	15.8	9.8	2.2	-4.3	(33)
(34)		MCDWB	-2.5	2.7	10.1	18.6	23.1	25.1	27.2	25.8	22.3	15.9	7.4	-0.6		(34)
(35)	Mean Daily Temperature Range		MDBR	11.5	12.2	12.2	12.6	12.4	11.0	10.1	10.7	12.5	12.2	11.1	10.9	(35)
(36)		5% DB	MCDBR	13.7	15.3	16.2	16.2	16.0	14.2	12.7	12.9	15.3	15.5	14.3	12.9	(36)
(37)		MCWBR	10.1	10.3	9.0	7.5	6.0	4.5	4.0	4.6	5.7	7.4	8.6	9.2	(37)	
(38)		5% WB	MCDBR	13.6	14.8	15.4	14.2	13.0	11.0	10.3	10.4	12.1	13.4	13.7	12.9	(38)
(39)		MCWBR	10.2	10.1	8.6	7.2	5.6	4.1	4.1	4.4	5.3	7.1	8.5	9.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.291	0.344	0.392	0.465	0.527	0.584	0.546	0.501	0.451	0.404	0.329	0.298	(40)	
(41)		taud	2.298	2.138	2.035	1.895	1.798	1.751	1.904	1.996	2.040	2.072	2.224	2.266	(41)	
(42)		Ebn at Noon	846	840	839	803	763	720	744	764	771	756	781	795	(42)	
(43)		Edh at Noon	86	120	151	187	212	223	190	168	149	127	92	82	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.53	3.51	4.69	5.59	6.18	5.73	5.58	5.40	4.68	3.58	2.55	2.16	(44)	
(45)		RadStd	0.15	0.21	0.29	0.34	0.40	0.45	0.33	0.31	0.25	0.22	0.18	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	-1.25	N/A	-0.36	-0.50	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	-1.31	N/A	N/A	N/A	N/A	N/A	+57	N/A

Nomenclature: See separate page